PANTONE REPORT : THE PAGES OF MICHEL DAVID

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A FEW WORDS ON THE INSTALLATION OF A PANTONE REACTOR ON A DIESEL ENGINE.

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ONE OF THE PROBLEMS THAT ARISE

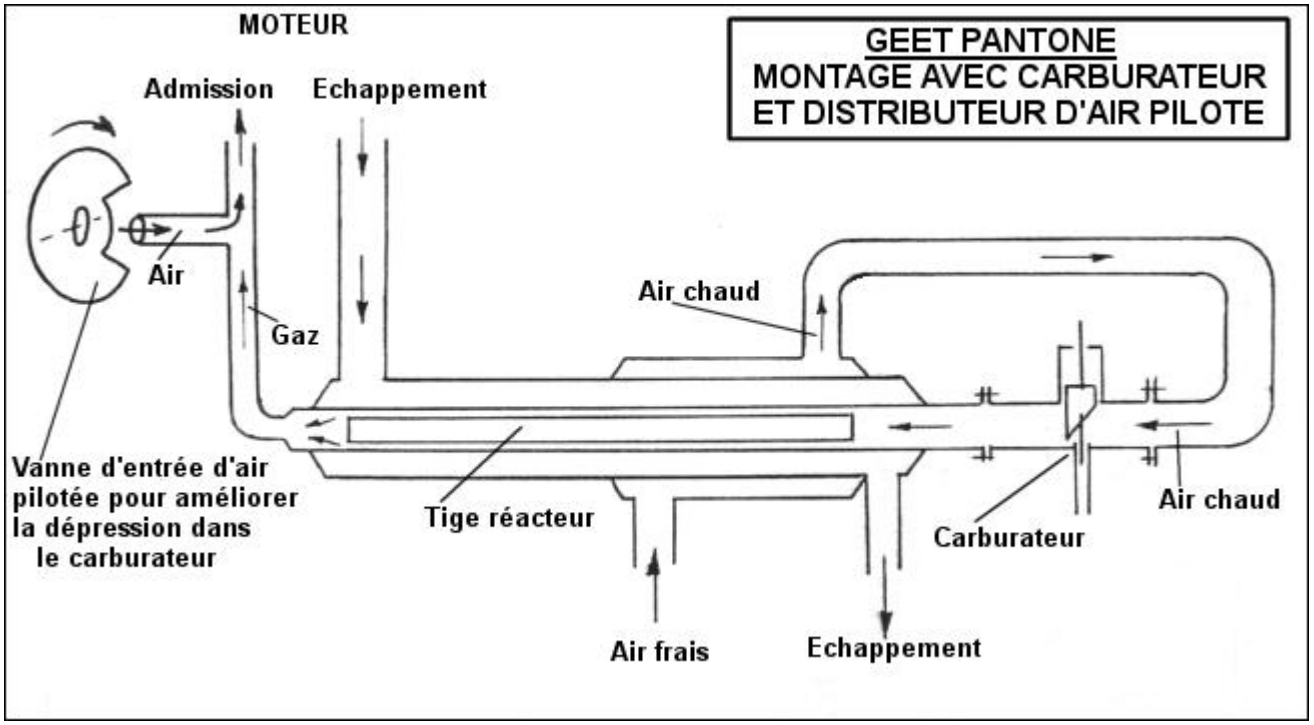
When one wants to install a GEET Pantone reactor on an engine and that one uses a carburetor or a bubbler with

bubbling by the ambient air and not by the exhaust gases, one quickly realizes the difficulty that both have to be operated as soon as the engine speed is a little low. Spraying is difficult to obtain satisfactorily in the small carburetor, because the vacuum is very low in the nozzle of the latter, it is the same for bubbling, the vacuum in the bubbler is insufficient to allow the air to flow. overcoming the pressure of fuel pitch. This problem, due to the presence of the air inlet valve, makes the operation of the installation very uncertain and if you want to do something serious, you have to find a correct solution.

In fact, the solution would be to be able to take turns sucking the gas from the reactor and then the air, instead of doing it at the same time as in all the assemblies described up to now. Thus when the gas would be sucked through the reactor, the vacuum would be correct in the carburetor or in the bubbler at all speeds.

In other words, when the piston goes down in the cylinder during the suction time, it would be necessary that at the beginning of the descent the air inlet valve is closed and that only the gas from the GEET is sucked and that it opens. then to let in the air necessary for the formation of the detonating mixture until the complete descent of the piston.

You immediately see that we are going to be confronted with mechanical problems since we will have to consider the manufacture of a mechanical shutter of the rotary type, to ensure this function, as well as its connection to the motor.



1- From top dead center to position 2, the suction is done through the reactor

2- Here only the air is sucked in, the reactor is no longer under vacuum

3- Bottom dead center before new start of cycle 1 and 2

4- Air-gas mixture valve to control the engine speed

The 240 cm3 engine running on fuel oil with distributor and carburetor

Despite everything, in the context of an exploitable and non-experimental realization as on an automobile where the engine must be able to function correctly at all the speeds, this is worth it because the good proportions of the explosive mixture can be obtained correctly at all the speeds. engine and it will even be possible for the device to be provided with a double centrifugal and vacuum adjustment system.

This device, which is a little complex to produce, is beyond the reach of the average handyman without machine tools. I will therefore not embark on a detailed description of such a device, the design of which may vary depending on the imagination of one or the other. For those who are more electronic than mechanics, it is possible to envisage a construction of the device based on an electromechanical valve and sensors on one of the motor pulleys, but it is still necessary to make this kind of valve controlled by an electromagnet capable of operating. at engine speed.

I considered and experimented with another way to overcome this problem of the air inlet valve between the engine and the reactor. I simply deleted this one and I connected the engine inlet directly to the outlet of a reactor that I designed with a larger gas passage section, identical to the section of the carburetor nozzle original engine reused in this case. In this adaptation, which is really the simplest that can be made from GEET PANTONE, all the air necessary for the constitution of the detonating mixture, passes into the reactor and is therefore no longer admitted directly into the engine and is used entirely. normal operation of the carburetor as if the reactor was not between it and the engine. This is quite suitable for small engines where each cylinder does not exceed too much 300 cm3,because beyond that, the size of the reactor would quickly be too large and it would be necessary to combine several reactor tubes to obtain a sufficient gas passage section, given that the space between rod and tube must not be exceeded too much 1.5 mm. With all these arrangements, it is in the interest of preheating the air, because the latter, in too large a quantity, tends to prevent the reactor from rising in temperature, which is not very good.

So, this assembly is interesting for all small lawn mower engines or others because it is not so possible to embark on the adaptation of a rotary air intake valve.

NOTE: all these carburettor-based experiments were done only with gasoline, but I am considering trying the addition of a second carburetor in series with the first to try to fuel water with the addition of a little soluble oil to prevent rusting of the small steel parts of it.

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